

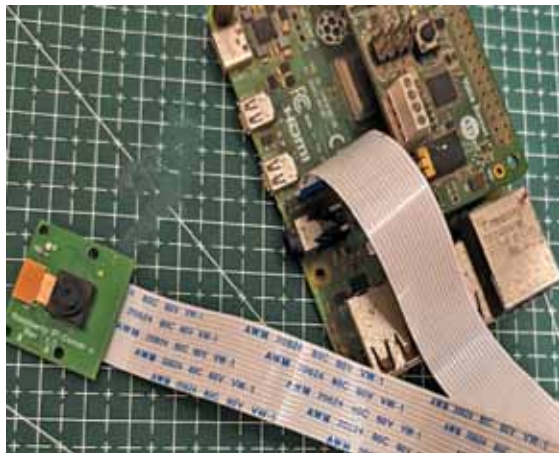


Fig. 6: Raspberry Pi connected with a camera

detected to ensure process compliance, verify tool availability, and prevent missing-tool incidents.

6. Edge AI prototyping. Edge AI applications can be developed and validated on resource-constrained platforms such as Raspberry Pi or low-power laptops, where compact models, high inference speed, and ease of deployment are essential.

These examples demonstrate how YOLOE enables rapid customisation of object detection models, making it particularly suitable for lightweight, task-specific edge AI applications that require minimal development effort and do not require extensive model retraining.

Connection

The system can be executed on a laptop, such as a MacBook Air, or on a Raspberry Pi 5 with 8GB RAM. For Raspberry Pi operation, the camera should be connected to the CSI camera port, and the board should be powered using a suitable AC-DC

adaptor. Fig. 6 shows the Raspberry Pi connected to the camera.

To access the Raspberry Pi desktop, VNC or another remote desktop application may be used. Alternatively, an HDMI monitor, keyboard, and mouse can be connected directly to the Raspberry Pi.

Testing

When the object detection program is executed,

the application automatically opens the connected camera and begins capturing live video. Each frame is processed in real time, and the detected objects are displayed with their class ID, object name, and bounding box. The annotated video stream appears on the screen, allowing the detection results to be verified immediately.

Aftermath

Object detection and recognition have traditionally required large datasets, lengthy training cycles, and high-end GPU hardware. The emergence of prompt-based vision models such as YOLOE has significantly reduced this barrier.

This system demonstrates that YOLOE enables custom objects to be added using simple text prompts or just a few reference images without extensive retraining or complex data preparation pipelines. Tasks that previously required days of data collection and training can now be

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completed in minutes.

This approach creates practical opportunities for hobbyists, students, researchers, and engineers to explore computer vision on everyday hardware, including laptops and single-board computers such as the Raspberry Pi. By focusing only on the required objects, the exported models are smaller, faster, and better suited for edge AI deployment while still delivering reliable detection performance.

Beyond demonstrating the capabilities of YOLOE, this system provides a practical framework for developing lightweight, task-specific vision applications in areas such as assistive technology, personal automation, smart homes, industrial monitoring, and education. **EFY**

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